

ASTRAKHAN, Russian Federation

WMO: 348800

Lat: 46.283N

Lon: 47.983E

Elev: -21

StdP: 101.57

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.1	-21.9	0.5	-17.7	-18.6	0.7	-14.5	11.9	-4.7	10.4	-2.9	2.9	60	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	36.9	20.8	35.1	20.4	33.7	20.1	23.5	30.3	22.5	29.8	21.8	29.3	4.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.2	26.3	20.3	14.9	25.4	19.3	14.0	24.9	69.9	30.0	66.2	29.8	63.4	29.2	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.4	8.3	DB	-21.0	38.8	5.3	2.1	-24.8	40.3	-27.9	41.6	-30.9	42.7	-34.7	44.3	(4)
(5)			WB	-21.2	24.9	5.1	1.1	-24.9	25.7	-27.9	26.3	-30.8	26.9	-34.5	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-3.8	-2.6	3.7	11.6	18.7	24.0	26.4	25.3	18.5	11.2	3.6	-1.5	(6)	
	DBStd	11.62	5.47	5.70	4.05	4.23	4.07	3.51	2.71	3.35	3.96	4.55	5.07	4.93	(7)	
	HDD10.0	1602	428	354	196	31	1	0	0	0	1	41	196	355	(8)	
	HDD18.3	3302	687	587	452	205	46	3	0	1	44	222	441	613	(9)	
	CDD10.0	2094	0	0	2	78	272	419	508	474	257	79	6	0	(10)	
	CDD18.3	752	0	0	0	3	58	172	249	216	51	2	0	0	(11)	
	CDH23.3	8384	0	0	0	46	592	1904	2914	2445	459	24	0	0	(12)	
	CDH26.7	3982	0	0	0	6	194	904	1501	1238	136	3	0	0	(13)	
(14) Wind	WSAvg	3.9	4.0	4.2	4.4	4.4	4.1	3.8	3.5	3.5	3.8	3.7	3.7	4.1	(14)	
(15) Precipitation	PrecAvg	215	15	13	16	21	25	24	17	18	21	18	17	17	(15)	
	PrecMax	375	41	52	64	87	113	80	81	79	69	75	44	43	(16)	
	PrecMin	76	1	0	0	0	0	0	0	0	0	0	1	0	(17)	
	PrecStd	64	10	10	13	20	22	20	17	19	17	17	11	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.0	17.2	27.5	33.2	38.0	39.0	39.0	33.1	26.1	17.2	9.9	(19)	
		MCWB	6.1	6.3	10.1	15.7	18.5	21.0	21.2	20.5	19.7	16.9	12.3	7.8	(20)	
	2%	DB	5.2	7.5	14.5	24.4	30.9	35.5	36.8	36.8	30.0	23.2	14.2	7.8	(21)	
		MCWB	4.0	4.7	8.3	14.3	18.0	20.2	21.0	20.7	18.3	14.6	10.9	6.2	(22)	
	5%	DB	3.9	5.8	12.2	21.9	28.9	33.9	35.1	34.8	28.0	20.9	12.2	6.0	(23)	
		MCWB	2.8	3.7	7.0	13.2	17.2	19.8	20.6	20.4	17.6	13.8	9.8	4.9	(24)	
	10%	DB	2.4	4.0	10.2	19.8	26.8	31.8	33.6	32.9	26.0	18.2	10.3	4.5	(25)	
		MCWB	1.5	2.5	5.9	12.2	16.6	19.3	20.3	20.2	16.6	13.1	8.5	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	6.8	10.8	16.7	20.6	24.4	24.7	24.7	22.2	18.1	13.1	8.1	(27)	
		MCDB	7.2	8.8	16.9	25.4	28.5	31.6	31.2	31.1	28.8	22.9	16.0	9.2	(28)	
	2%	WB	4.3	5.1	9.0	15.2	19.4	22.6	23.2	23.5	20.5	16.4	11.5	6.2	(29)	
		MCDB	5.1	6.7	13.4	22.4	27.1	30.1	30.3	30.0	26.3	20.9	13.7	7.1	(30)	
	5%	WB	2.8	3.8	7.6	14.0	18.5	21.5	22.5	22.5	19.2	14.9	10.3	4.8	(31)	
		MCDB	3.6	5.3	11.3	20.5	25.8	28.9	30.0	29.5	25.1	18.7	12.0	5.6	(32)	
	10%	WB	1.5	2.6	6.4	12.7	17.6	20.6	21.7	21.7	18.1	13.6	8.7	3.5	(33)	
		MCDB	2.3	3.8	9.6	18.2	24.4	28.1	29.6	29.1	23.7	17.6	10.3	4.3	(34)	
(35) Mean Daily Temperature Range	MDBR		6.3	7.9	9.3	11.1	11.3	11.9	12.2	12.7	11.7	10.2	7.4	5.5	(35)	
	5% DB	MCDBR	6.7	9.4	12.5	14.6	13.9	14.7	14.6	15.2	14.0	13.3	9.6	6.6	(36)	
		MCWBR	5.4	6.9	7.2	6.4	4.6	4.5	4.2	4.8	5.1	6.2	6.5	5.4	(37)	
	5% WB	MCDBR	6.3	8.6	11.5	13.4	12.2	13.0	12.9	13.4	12.2	11.5	8.8	6.2	(38)	
		MCWBR	5.3	6.5	7.1	6.3	4.8	4.5	4.4	5.1	5.4	6.5	6.3	5.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.362	0.390	0.427	0.494	0.462	0.445	0.451	0.456	0.428	0.393	0.371	0.374	(40)	
	taud		2.143	2.079	2.063	1.936	2.105	2.206	2.201	2.163	2.183	2.237	2.207	2.111	(41)	
	Ebn at Noon		688	750	783	764	810	826	815	792	777	744	677	618	(42)	
	Edh at Noon		93	120	141	176	154	140	139	139	124	102	87	87	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.06	1.94	3.33	4.92	6.19	6.80	6.56	5.87	4.47	2.74	1.40	0.81	(44)	
	RadStd		0.16	0.28	0.36	0.43	0.48	0.44	0.33	0.29	0.31	0.28	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.57	N/A	N/A	+1.50	N/A	N/A	N/A	N/A	+170	+115	(46)	
(47) Regional (2 neighbors)		+0.53	N/A	N/A	+1.46	N/A	N/A	N/A	N/A	+170	+115	(47)	

Nomenclature: See separate page